

VIETNAM NATIONAL UNIVERSITY-HO CHI MINH CITY

HO CHI MINH UNIVERSITY OF SCIENCE

FACULTY OF INFORMATION TECHNOLOGY

KNOWLEDGE ENGINEERING DEPARTMENT

Homework 4

Course: Introduction to Artificial Intelligence

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Contents

1 Problem 1: Hill Climbing Algorithm	2
1.1 Which places in the state space make regular hill climbing fail to reach the optimal solution?	2
1.2 Are variations of hill climbing guaranteed to be complete and/or optimal? Briefly explain.	2
2 Problem 2: Minimax with Alpha-Beta Pruning	3
2.1 Fill in each blank circle with utility values.	3
2.2 What is the best action for MAX?	3
2.3 Branches pruned by Alpha-Beta Pruning (left-to-right)	3
3 Problem 3: Stochastic Game Tree (Expectimax)	4
3.1 Fill in each blank circle with utility values	4
3.2 What is the best action for MAX?	4

1 Problem 1: Hill Climbing Algorithm

1.1 Which places in the state space make regular hill climbing fail to reach the optimal solution?

Regular hill climbing fails to reach the optimal solution in the following situations:

1. **Local Maxima:** This is the most common failure point. A local maximum is a solution that is better than its neighbors but not the best overall. Since the algorithm only considers neighboring solutions, it cannot move away from this local peak to find a potentially better global maximum. Hill climbing gets stuck here because it only moves to states with higher values, and once it reaches a local maximum, there are no adjacent states that offer improvement.
2. **Plateaus (Flat Local Maxima):** A plateau is a flat area in the state space where all neighboring states have the same value as the current state. Hill climbing cannot make progress on a plateau because there's no clear "uphill" direction to move towards. This can lead to stagnation, where the algorithm cannot make progress.
3. **Ridges:** In cases where the optimal solution lies on a ridge, the algorithm may fail to navigate the steep slopes effectively. It may move down the ridge instead of following it upward toward the optimum. A ridge is a sequence of local maxima that are not directly connected by an ascending path. The algorithm might move along a ridge, but it can get stuck if the path to the global optimum requires moving "downhill" to reach another part of the ridge or an entirely new region of the state space.

1.2 Are variations of hill climbing guaranteed to be complete and/or optimal? Briefly explain.

No, variations of hill climbing do not guarantee completeness or optimality:

- **Random-restart Hill Climbing:**

- *Completeness:* Not guaranteed. May still get stuck in local optima even with multiple restarts.
- *Optimality:* Not guaranteed. Each restart still uses basic hill climbing, which can fail at local optima.

However, it increases the probability of finding the global optimum by trying multiple starting points.

- **Local Beam Search:**

- *Completeness:* Not guaranteed. All beams might converge to the same local optimum.
- *Optimality:* Not guaranteed. Still susceptible to local optima, though less likely with multiple search paths.

Maintains multiple states simultaneously, but can suffer from lack of diversity.

Reason: These are still local search algorithms that make decisions based only on local information. They cannot guarantee finding the global optimum in spaces with multiple local optima.

3 Problem 3: Stochastic Game Tree (Expectimax)

3.1 Fill in each blank circle with utility values

Here's the tree with the filled-in utility values for the stochastic game:

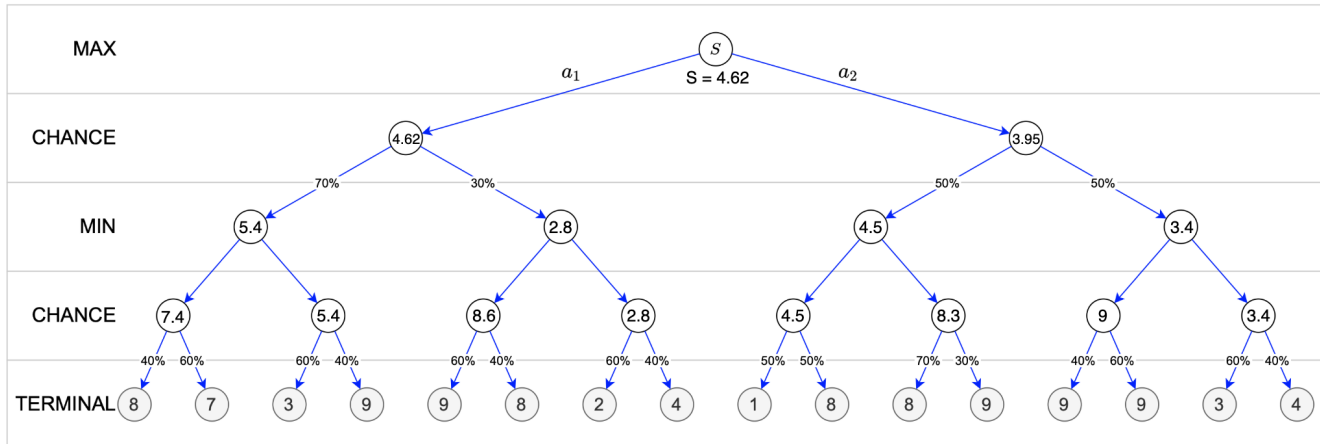


Figure 3: Stochastic Game Tree with Expectiminimax Utility Values

3.2 What is the best action for MAX?

Based on the Expectiminimax evaluation, the utility value of the root node S is 4.62. This value is achieved by choosing the left branch (action a_1) from the root, which leads to a CHANCE node with an expected value of 4.62. Therefore, the best action for **MAX** is to choose a_1 .